Supplementary information: modeling electrical losses in CsPbI3 solar cells

Hussain Touhid Siddiquee Md. Abdul Malek Fahim

This supplement carries the audit trail removed from the main paper: receipt inventory, script verification, extended screening detail, optical diagnostics, prior accounting, and reference notes.

S1. Receipt inventory and run accounting

The main paper reports evaluated values; this section records how they were produced. The 96-run grid had six worker-file misses. The 36-run refinement had three misses. The prior-volume design had 95 converged in-prior runs, eight separate $C_r = 0$ sensitivity runs, and one recorded solver divergence. The density-of-states extension had 96 in-prior runs and eight sensitivity runs, with 104 converged. Retries were solo reruns after worker crashes or hangs; they are recorded in the receipts rather than silently excluded. Copyrighted reference PDFs are listed in the repository, not redistributed.

Run set	Runs (converged)	Key output	Revision `6e991c1`
Grid Round A	96 (93 usable)	`outputs/roundA.jsonl`, `receipts/receipt_roundA.json`	archived
Grid Round B refinement	36 (33 usable)	`outputs/roundB.jsonl`, `receipts/receipt_roundB.json`	archived
Prior-volume envelope	96 (95) + 8 $C_r = 0$ + 1 divergence	`outputs/s25_morris_summary.json`-class analysis JSONs	archived
Density-of-states extension	96 + 8 sensitivity (104)	`outputs/s26_diag.json`, `outputs/s31_small.json`	archived
Series-resistance sweep	8 + 3 extension	`outputs/s36_rs.json`, `outputs/s1_dose.json`	archived
Joint affinity map	52	`outputs/s11_joint_chi.json`	archived
Forward-check calibration	2 films	`outputs/s35_phys.json` (17.96%/16.76%)	archived
Analytic bounds	3 gap steps	`outputs/s18_analytic.json` (24.4/23.4/19.8 mA/cm^2)	archived

Table 1: Receipt inventory for the archived runs. Misses and divergences are recorded in the receipts rather than silently excluded.

Key receipts are: baseline validation; grid rounds A and B; characterisation and audit runs; interface dose, resistance (including the Rs=3–5 extension in `outputs/s36_rs.json` plus the 1.70 eV ceiling variant at 24.73%), and work-function runs; joint-affinity maps; radiative and mobility pricing; stacked realistic variants; calibration ladder plus the vapour/spin forward check (`outputs/s35_phys.json`: 17.96%/16.76%); prior-volume and density-of-states extensions; contact and doping interaction checks; analytic bound calculations (independent step-absorption Jsc 24.4/23.4/19.8 at 1.65/1.70/1.80 eV in `outputs/s18_analytic.json`); and automated screens over archived receipts. The density-of-states record-comparison rows are labelled by run id: J_dos03_cr10 (23.46%, $N_v = 4.1 \times 10^{18}$, $C_r = 10^{-10}$), J_dos05_cr10 (23.04%, $N_v = 8.9 \times 10^{18}$, $C_r = 10^{-10}$), J_dos03/J_dos05 without C_r (26.50%/26.11%, screen-excluded).

S2. Script verification and unit traps

SCAPS definition files store SI units while script commands use practical units. A same-value check is therefore required before interpreting any scripted defect, doping, mobility, or radiative change. Setting absorber doping to 10^{15} cm^{-3} reproduces the file baseline, while an SI-valued input injects a million-fold overdose and creates ghost cells.

A second trap concerns interface cross-sections: the script key for interface capture cross-section restructured the interface defect block in this campaign rather than scaling the addressed rate. The definition-file route reproduced the expected $N \times 10$ result, while the script-key route did not. The working rule is to verify every defect-population change with a same-field query in the same script and to treat a wrong-direction efficiency move as a failed application.

A near-zero radiative test at 10^{-16} leaves the baseline unchanged to 0.0001 points. That establishes omission as below solver sensitivity for this comparison; no unavailable external equation is cited for the built-in radiative estimate.

S3. Extended screening detail

The full Round A and Round B leaderboards, joint-affinity extensions, gap steps, doping–affinity checks, and contact-regime variants are audit detail. Their headline-relevant conclusions are retained in the main paper: the landscape is converged rather than climbing; the joint-affinity peak is interior rather than edge-sitting; gap steps track the activation-energy landscape as well as absorption; and contact tolerance collapses as affinities move to aggressive values.

The no-interface audit points are reported as disqualified by lack of radiative margin. The $N\sigma$ identities are retained as verification checks: symmetric interface scaling and bulk scaling reproduce the corresponding density-scaled runs, confirming that the tested rate scales with the intended product.

S4. Optical and current diagnostics

The QE integral exceeds the converged IV current by 1.19 mA cm^{-2}: 22.39 versus 21.20. The IV value reproduces across IV step sizes, while its internal generation/recombination integral gives 21.21. The QE curve is also identical under the tested illumination modes. Ideal step absorption gives 24.4 at 1.65 eV for the archived spectrum. After the stated 10% front reflection, the corresponding ceiling is 21.96, below the QE integral. The IV current is therefore retained.

Spectrum file substitution through the script interface returned the default-spectrum result digit-identically, so independent spectrum discrimination was blocked rather than passed. The residual QE-path over-delivery is carried as a two-sided ± 1.2 mA/cm^2 absolute-scale systematic. A separate blue-response uncertainty of order 1 mA/cm^2 arises from absent dispersive parasitic absorption. These are stated as uncertainties, not corrections.

Blocked test	Attempted route	Result / proposed resolution
Spectrum discrimination	Script-interface spectrum-file swap	Script returned the default-spectrum result digit-identically; blocked. Manual SCAPS GUI run with the spectrum file selected is the proposed resolution.

Table 2: Blocked diagnostics recorded for the optical audit trail.

Capacitance–voltage, Mott–Schottky, band-diagram, and generation/recombination outputs are solver self-consistency checks. The low-bias Mott–Schottky branch recovers the input doping, and

the biased profiles locate the ceiling voltage deficit at back-interface recombination. They do not independently validate the device.

S5. Prior-volume accounting

The headline envelope uses only the 95 converged in-prior runs: median 17.98%, p5 15.55%, p95 19.96%, rounded to 18.0% and 15.6–20.0%. The eight $C_r = 0$ runs have median 17.93% and are reported separately. One edge-contact configuration diverged repeatedly and is recorded as non-convergent, not excluded as a crash.

The density-of-states extension gives median 18.18%, p5 15.76%, and p95 20.03%, rounded to 18.2% and 15.8–20.0%. Its density-of-states rank importance is near zero in-volume, even though physical density of states swings the clean corner by more than a point. Corner sensitivity and volume sensitivity are therefore different statements.

Bulk and interface quality were sampled independently. Binning by the nominal interface-to-bulk velocity ratio does not reveal a clean internal crossing at this sample size. The ratio organises the two anchor regimes but does not resolve the ranking bin by bin.

S6. Calibration limits

The measured-grade ladder climbs monotonically and saturates at 20.32–20.66%, below the certified record. Published device tables do not provide enough defect-level detail to reconstruct portable bulk lifetime and interface velocity for a specific device. The ladder is therefore a transfer-error bound, not a device-specific hindcast. No posterior calibration is claimed because no device data enter the prior loop. The vapour/spin forward check (260 nm films, N_t from reported lifetimes: 17.96% vapour at $N_t = 10^{13}$, 16.76% spin at $N_t = 5 \times 10^{14}$) recovers the correct order and plausible scale without fitting. The 22.02% certified value (Dai et al.) is taken as the highest certified CsPbI3 value among the cited records (next certified: 21.43% inverted, 20.35% carbon HTL-free); the 22.16% and 22.15% laboratory values are carried for completeness but excluded from the certified ranking. The Xu et al. 22.15% entry remains bibliographically unverified and is flagged as such in the bibliography.

S7. Protocol mapping and reference notes

The main-paper checklist maps to realistic inputs, full reporting, effective spectra/absorption, SQ screening, and explained deviations. The optical transfer-matrix half is cited rather than reimplemented [1]; the electrical half is implemented here. The broader survey diagnosis is cited for inflated SCAPS results and their causes [2].

Shah et al. [3] is the first-author citation for the 20.6% buried-interface result; a timeline label using only the second author's name has been corrected to Shah. Cited experimental values are taken as reported in their sources. The Zhang et al. [4] record is cited without a DOI because no verified DOI was available in the local bibliography. The Dai et al. [5] laboratory/certified distinction is retained as 22.60% laboratory and 22.02% certified.

Supplementary References

[1] S. Guruprasad, A. Malik, A. Saidarsan, P. Basumatary, and D. S. Ghosh. Modeling optical losses in perovskite solar cells: A modified framework toward realistic performance projection. *Solar Energy*, 305:114251, 2026. doi: 10.1016/j.solener.2025.114251.

[2] A. Saidarsan, S. Guruprasad, A. Malik, P. Basumatary, and D. S. Ghosh. A critical review of unrealistic results in SCAPS-1D simulations: Causes, practical solutions and roadmap ahead. *Solar Energy Materials and Solar Cells*, 279:113230, 2025. doi: 10.1016/j.solmat.2024.113230.

[3] S. F. A. Shah, I. Jeong, J. Park, D. Shin, I. Hwang, N. Tsvetkov, D. Kim, J. Gwak, J. H. Park, S. I. Seok, K. Kim, and H. Min. Efficient and stable CsPbI3 perovskite solar cells with spontaneously formed 2D-Cs2PbI2Cl2 at the buried interface. *Cell Reports Physical Science*, 5:101935, 2024. doi: 10.1016/j.xcrp.2024.101935.